Practical Lab Week 2

Objective: the objective of this lab is to practice branching in Java.

1. Write a Java program that asks the user to enter a traffic light colour (red, yellow, or green) and then prints an appropriate action (e.g., "Stop", "Slow down", "Go").
   Example program output:
   ```
   Enter traffic light colour (red, yellow, green): red
   Action: Stop!
   Enter traffic light color (red, yellow, green): yellow
   Action: Slow down!
   Enter traffic light color (red, yellow, green): green
   Action: Go!
   Enter traffic light color (red, yellow, green): blue
   Invalid input! Please enter red, yellow, or green.
   ```

2. Write a Java program that asks the user to enter a password and evaluates its strength based on the following rules:
   a. Weak: Less than 6 characters OR contains only letters OR contains only digits.
   b. Strong: More than 10 characters AND contains letters, digits, and at least one special character (!@#$%^&*).
   c. Moderate: All other cases.

   ```
   Enter a password: hello
   Password Strength: Weak

   Enter a password: password123
   Password Strength: Moderate

   Enter a password: Str0ng@2024
   Password Strength: Strong
   ```

3. Write a Java program that takes a math expression as a String (e.g., "10 + 5") and evaluates it. Hint: Ask the user to enter the expression in 'number operator number' format with space between the number and operator. As such, you can split the user input using “space” to parse the number and the operator.

   ```
   Enter an expression (e.g., 10 + 5): 10 + 5
   Result: 15.0
   Enter an expression (e.g., 10 + 5): 10+5
   Invalid format! Please enter in 'number operator number' format with space
   between the number and operator.
   Enter an expression: 20 - 7
   Result: 13.0
   Enter an expression: 8 * 4
   Result: 32.0
   Enter an expression: 9 / 2
   Result: 4.5
   Enter an expression: 10 $ 5
   Invalid operator! Please use +, -, *, or /.
   ```

4. Write a Java program that asks the user for their birth month and day and determines their zodiac sign.
   Hint: You can find a list of all Zodiac signs & their Unicode codes in the following link: https://en.wikipedia.org/wiki/Astrological_symbols

```
Enter your birth month (1-12): 12
Enter your birth day (1-31): 30
Your zodiac sign is: Capricorn ♑
```